

A Tool for Guiding Teachers and their Learners: the Case Study of an Art Class

Rosella Gennari
Free University of Bozen-Bolzano
Bolzano, Italy

Alessandra Melonio
Ca' Foscari University of Venice.
Venezia, Italy

Mehdi Rizvi
Heriot-Watt University
Edinburgh, United Kingdom

ABSTRACT

There is growing interest in how to enable in-service teachers to promote computational skills in their classes. Different researchers also highlight the importance of developing a critical mindset toward digital technology. This paper reports a case study, part of a training program for in-service teachers, organised by researchers and the local School District, to bring physical computing into class, and foster critical reflections on technology along the process. The program included teachers for diverse subjects, across primary and high schools. It used the novel IoTgo tool, adaptable to the needs of the different teachers. The study in this paper considers the case of an art class in a high school. According to the reported data analysis, the tool and the structure it offered enabled the teacher to bring computing into her art class, on her own, and learners to reflect along the process from different perspectives.

KEYWORDS

IoT, smart, reflection, teacher, learner, independence, phygital, tool

ACM Reference Format:

Rosella Gennari, Alessandra Melonio, and Mehdi Rizvi. 2023. A Tool for Guiding Teachers and their Learners: the Case Study of an Art Class. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3544549.3573863>

1 INTRODUCTION AND RELATED WORK

Smart things, such as smartphones, are pervasive; people, especially teens, are surrounded by them. They exchange data through physical devices, i.e., input devices such as sensors, and output devices acting as actuators. Examples are a motion sensor as input for the movements of people around, and an LED matrix for light effects as output. A microcontroller or another similar computing board can be embedded in a physical thing, such as a lamp, and make it “smart” with a suitable program for the behaviour of its input and output devices. In the case of the lamp, the program can use the motion sensors to detect whether there are people around and hence light up the LED matrix in bright colours.

Various tools include cards for enabling non-experts to design smart things. For example, the Tiles IoT Inventor Toolkit has different cards for different environments [12]. In the IoT Design Kit, flashcards help convey design expectations [4]. Recent tools for

children or teens couple cards with micro-controllers or similar boards to prototype ideas of smart things, e.g., SNaP uses micro:bit boards [5, 6, 11]. Such work is in line with past research which tries bringing to the table the concreteness in building technology and a new expressive media for motivating younger generations to create technology-enhanced artefacts [8]. Lately, how to promote critical reflections on technology has also become a crucial concern for several researchers working with non-experts, and especially children, in digital technology design [10, 13].

However, this and other similar tools are used by design researchers, professionals, or practitioners in technology labs and are not generally adopted by teachers in different grades and across disciplines to teach computing skills yet. To the best of our knowledge, there are few examples of integrated learning experiences, mainly in primary schools and in North Europe or the USA, in which teachers teach different subjects and promote computing skills without experts’ interventions, e.g., [9]. As Angeli and Giannakos point out in their editorial, what is currently missing is how computing skills “might map to different abilities, grade level, disciplines, gender, and educational level”; to reach it, “teachers need to be systematically prepared” and “professional development programs need to be implemented for in-service teachers” [1]. This paper intersects such concerns.

Starting in Autumn 2021, a course was organised with the local Department for Schools for in-service teachers from primary to high schools, all teaching different subjects. The course trained teachers to physical computing with a problem-based, design-oriented perspective [2]. It adopted the IoTgo tool for schools, for the first time presented in this paper, that evolved from past versions for creating with artists novel art-artefacts [7]. The tool is designed to be adaptable to different grades, skills, and subjects. It combines physical, paper-based game cards and boards, with a digital app and off-the-shelf physical-computing devices, namely, micro:bit boards, physical input devices (e.g., buttons, motion sensors), and output devices (e.g., LED strips, motors). In Spring 2022, several teachers used IoTgo for organising their own activities at school, adopting physical computing for their own subject.

This paper reports the case study of an art teacher from high school, organising a STEAM activity for her class with IoTgo for schools, combining art and physical computing. Qualitative data were gathered to gain a holistic perspective that retains the integrity of the case, through a cross-case analysis of the relevant findings [3]. Specifically, the teacher kept a diary of activities at school and discussed with researchers the benefits and pain points of her experience with IoTgo for her own class. Learners’ voices were also accounted for, considering what they managed to create and reflect on with IoTgo, and documented through it. The paper starts describing the IoTgo tool used by the teacher and her class.

Then it reports the case study, and finally it discusses its main findings.

2 IOTGO FOR AN ART CLASS

IoTgo consists of paper-based cards and plug-ins, three game boards, micro:bit physical-computing boards, and a dedicated app. The second and third boards embed a reflection area, which stirs reflections and revisions on smart things. Each board is described separately in the following. The app and areas for reflecting are described subsequently.

2.1 The First Board for Creating a Problematic Context and Imagining Smart Things in the Wild

The initial board asks teens to start from a story excerpt and continue it, so as to narrate, together, problematic situations that matter to them, besides possible locations and people affected by the situations. Situations, locations, and people described by teens are then used by researchers to enrich the story and create related cards for the follow-up board, concerning the context. Moreover, this board asks learners to start imagining possible smart things for the story and chosen cards, which they can change or evolve with the other boards. See Fig. 1.

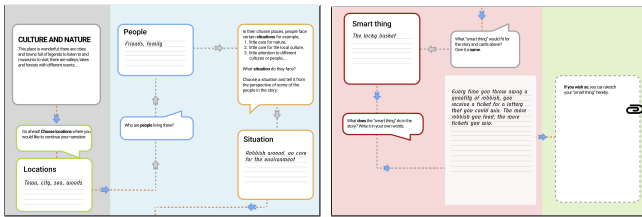


Figure 1: Two main areas of the first board: for co-creating IoTgo story elements (left); imagining a new smart thing “in the wild” for the story, without guidance (right).

2.2 The Second Board for Analysing the Context with Cards

The board for the context analysis asks teens to start from the enriched story, analyse and elaborate on the problematic situations in it, and hence choose a thing to make smart for it. Specifically, the board instructs them to choose a specific location where to place their thing to make smart, consider possible evocative smells therein, besides persona and a mission to solve with the thing for the context. See Fig. 2.

2.3 The Third Board for Ideating for the Context with Cards

The physical board asks teens to create a program for their smart thing. Specifically, it invites them to choose physical input and output cards (e.g., touch sensors, buttons, LEDs, speakers). Then it enables them to automatically generate a program for them with the companion IoTgo app, described in the following. See Fig. 3 for the physical board.

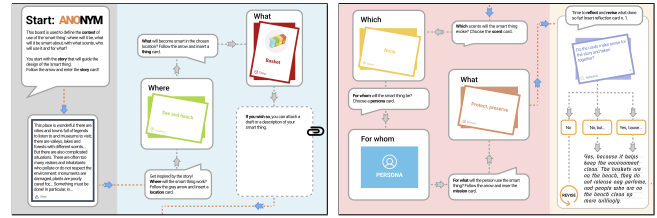


Figure 2: Two main areas of the second board of IoTgo for analysing the context with cards: choosing the location and thing to make smart for the story (left); deciding an evocative scent, for whom and for what the thing is, besides reflecting and revising (right).

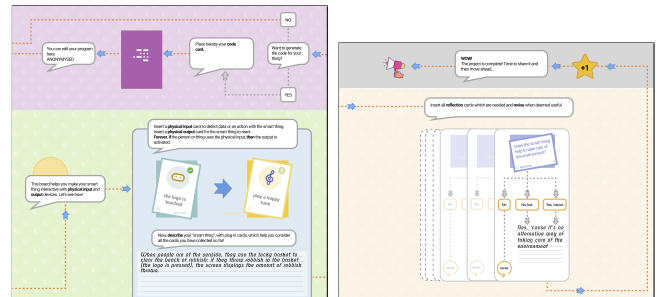


Figure 3: Two main areas of the third and final board of IoTgo: for ideating the smart thing with cards for physical input and output devices and generating a program with the app (left); reflecting and revising (right).

2.4 The App for Generating a Program for the Idea and Testing

IoTgo for schools comes with a web app that enables learners to generate automatically the program controlling the behaviours of the chosen combination of input and output cards, selected for their smart things. It generates programs in JavaScript and with MakeCode blocks. Learners can then further edit their program and upload it on micro:bit boards for prototyping their smart things. See a screenshot in Fig. 4. In this manner, learners can rapidly test their ideas of smart things in action, then reflect on them and revise them accordingly in the second or third boards.

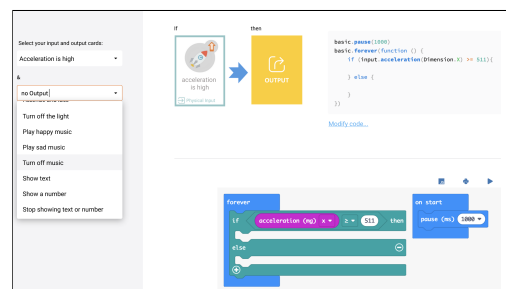


Figure 4: A screenshot of the IoTgo app.

2.5 The Cards and Plug-ins for Reflecting and Revising

Reflections are solicited with cards in relation to relevance in the context board, displayed in Fig. 2, and in relation to different topics in the physical board, shown in Fig. 3. These topics are as follows:

- (1) sustainability for the **environment**;
- (2) sustainability for **culture and people**;
- (3) **usability, feasibility and creativity**.

Reflection questions, printed on the reflection cards, ask to consider specific issues for a topic, e.g., a card displays the following reflection question in relation to the topic of sustainability for culture and people:

Is the smart thing safe for its users or others?

The reflection areas on the second and third boards ask learners to attach at least one reflection plug-in and card. The plug-in asks them to elaborate on the question on the card, and to answer in three possible manners:

- (1) “yes, ‘cause”, in case they answer in the positive to the reflection question, and why so;
- (2) “no, but”, in case the question is only partially tackled and they have to elaborate on reasons why it cannot be fully satisfied;
- (3) “no”, in case the question is answered in the negative and they need to revise their decisions.

Moreover, teens are invited to elaborate their own reflection questions, place them on empty reflection cards, and share them with peers. See Fig. 5.

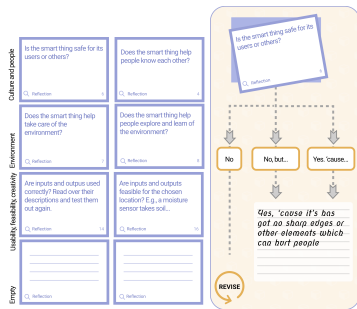


Figure 5: Examples of IoTgo reflection cards with questions per topic, and empty cards to create new reflection questions (left); reflection plug-in for the board, where learners insert a reflection card and their answers (right)

3 CASE STUDY

3.1 Context and Participants

The activity with IoTgo took place in the laboratory of a humanistic high school. It involved 13 teens, aged 14–15 years old (7 females, 8 males), attending the first year of their school, beside their art teacher. Before using IoTgo, learners had tried tutorials for micro:bit in MakeCode with their teacher. None of them had any programming experience except one who had used the Scratch block-based

programming environment for creating animations. Teens worked with IoTgo in pairs and, in one case, in a group of three.

The teacher organised her activity with IoTgo in 5 days, one day per week and for circa one hour and a half per day, towards the end of the school year. Researchers were present in class only when needed to introduce the material teens would use and available for the teacher in case she needed support.

Informed consent forms were distributed prior to the participation of the class in the activity, and data were processed in line with the recommendations of the ethics board of the university of researchers responsible for the study.

3.2 Leading Research Questions and Data Collection

The overall purpose of IoTgo is to make teachers as independent as possible in conducting their own classes, embedding physical computing in their discipline. The leading research questions of the case study were thus as follows:

- R1.** what IoTgo enables the art class and their teacher to do, without the support of researchers;
- R2.** what reflections it elicits, if any.

Since researchers were not at school, data were collected indirectly and with the following instruments to include the voices of all participants: (1) IoTgo boards filled in by learners; (2) diaries filled in by the teacher, coupled with interviews with the teacher.

Diaries were filled in by the teacher at the end of each class with IoTgo. They guided the teacher to reflect on the experience in class, with probing questions related to what was done by teens and their reflections. An example of what the teacher wrote on a diary page is in Table 1. In case of relevant issues, interviews with the teacher helped researchers clarify them. Boards of IoTgo enabled researchers to track learners' ideas over time, that is, from their conception from a story excerpt in the first board (Fig. 1) to their final evolution through the second and third boards (Fig. 2 and 3).

Questions	Answers
<i>IoTgo asked the class to analyse a given context with cards. How do you think it went?</i>	The class were very quick, mainly because they had already done some thinking (related to the context) in the first phase of the work.
<i>IoTgo asked the class to reflect on what they did. How do you think it went?</i>	They did a good reflection on the process followed to arrive at the thing to be made smart. One group realised that their smart thing was not relevant (for the context) and quickly redesigned a new one, thanks to the reflections they had made. They seemed more aware of the design process than in the initial experience, in which they had no steps and no tools for reflection (the first board).

Table 1: An excerpt of the diary for the teacher in relation to the second board: probing questions and answers

3.3 Results

Results are reported in relation to the aforementioned research questions. Data are processed together so as to gain a holistic perspective over the experience in class.

3.3.1 Smart Things. Table 2 shows the evolution of the descriptions of learners' ideas of smart things. All initial ideas were mostly maintained except in one case (4). However, all ideas changed considerably. According to the teacher, the guidance of the second and third boards was essential for maturing ideas: *“the only problems that emerged are not related to the organisation of the work, in fact, I believe that without this structure it would not have been possible to arrive at the final product in almost all groups”*.

Whereas the initial ideas tended to be abstract and generic, the final ideas tended to be all feasible to develop into working prototypes. In particular, all final ideas made clear what physical inputs and outputs were meant for, and how they were used. An example is given by the idea concerning a monument (5 in the table). Whereas the initial idea described a motion sensor capable of measuring a load, the final idea correctly used the motion sensor for detecting movement around and in relation to the chosen mission.

It is also worth noticing that the descriptions of the final ideas tended to follow similar patterns, close to the behaviour of the programs which the app generates, albeit with differences (e.g., different outputs). Two cases stand out, which were more complex and required variables for counting (1 and 5). These went beyond what the app currently generates and required the teacher's assistance with the initial support of a researcher. Moreover, the teacher reported that the creation of programs for prototypes, facilitated by the app (see Fig. 4), and rounds of revisions guided by reflection cards helped learners understand better and better what the physical-computing devices could do for their smart thing and they helped learners revise their ideas. Such revisions, guided by the board and teacher, required two days of the overall activity: *“This reflection (with the final board, input and output physical devices) was more complex than the previous ones because it raised very practical and concrete questions that were not obvious to them. This phase took up a lot of time and prevented some groups from sharing the final result”*.

3.3.2 Reflections. Reflections were also tracked on IoTgo boards via the dedicated plug-ins and cards (see Fig. 5). Interestingly, the 16 reflection cards that learners used were related to usability, feasibility, and creativity. Specifically, only one card for creativity was used, whereas all others were related to usability or feasibility. This is consistent with what was reported by the teacher, that is, that rounds of revisions were required mainly for the inputs and outputs in the final board, and using them properly.

Other 10 cards were used to reflect on environmental issues, that is, paying attention to waste in nature and preserving natural resources. Finally, 6 cards were related to culture, and mostly for monitoring crowds or avoiding damage to monuments. Example reflections are reported in Table 3 in relation to themes, questions, and ideas by learners.

Id.	Initial idea	Final idea
1	Every time you throw away a quantity of rubbish, you receive a ticket for a lottery that you could win. The more rubbish you feed, the more tickets you win.	When people are at the sea-side, they use the lucky basket to clear the beach of rubbish: if they throw rubbish in the basket (button pressed), the screen displays the amount of rubbish thrown.
2	When someone throws waste into the bin, the sensor plays a happy tune that invites people to continue throwing waste into the bin.	When the inhabitant is in the park, she uses the musical basket to protect the environment: if she throws the waste into the basket, then the (movement) sensor detects movement and the basket plays a happy tune.
3	When a person passes through an area with dry plants, a watering can draw his attention and sprays water for a certain amount of time.	When the plants are dry, there is little humidity, and then the water can sprinkle on them (with a motor).
4	Whenever people use harmful products on plants, the sensor detects them and purifies them.	When a tourist is in the museum, the decibel detector for Ötzi is used to maintain silence: if the noise in the museum is high, an alarm is played, inviting them to maintain silence.
5	A motion sensor detects when people sit on a monument and makes a noise, like an alarm, if it (the monument) is touched.	When a tourist is close to a monument, it uses 'micro 4 monuments' to see if they are too close to the monument (with a movement sensor); an alarm is then played and a message is heard telling them to move away and take care of the monument.
6	When a tourist is in a museum, he uses the bar to enter. The bar is used to monitor entries: when the limit of people in the museum has been exceeded, the bar closes and emits a bad smell.	When a tourist is in a museum, he uses the bar to enter. The bar is used to monitor and count the number of entries (with a tilt sensor): when the limit of people in the museum has been exceeded, the bar closes (with a motor) and emits a bad vanilla smell.

Table 2: Ideas by learners with their id(entifiers), and their evolution with IoTgo.

Id.	Theme	Question	Answer
2	Environment	Does the smart thing help take care of the environment?	Yes, because it is an alternative way of taking care of the environment
5	Culture	Does the smart thing help know, preserve or enhance the cultural heritage of the place where you live?	Yes, because it avoids people damage it
2	Feasibility	Can the inputs and outputs be used with the smart thing?	Yes, because they can be glued to the basket

Table 3: An example answer per reflection theme and question of IoTgo, besides id(entifier)s of ideas by learners.

4 DISCUSSION AND CONCLUSIONS

There is an increasing interest in teaching children and teens computational skills necessary for developing novel technologies, and in critically reflecting on these. However, recommendations concerning how to foster computational skills with a critical mindset are often given in general terms: this “presents challenges for teachers and educators who wish to develop learning activities for their students and make them shareable and adaptable to different contexts and needs” [13]. This paper tackled this and similar concerns. It presented the novel IoTgo tool, which was used in a training program for in-service teachers for different subjects, e.g., maths and art. Teachers used the tool to learn about physical computing, within a problem-based and design-oriented approach, and then they brought what they had learnt into their classes, for their own subject. This paper presents a case study of an art class at a high school, whose teacher adopted the tool for developing smart artefacts with physical-computing devices and enabling teens to critically reflect along the development.

The tool is divided into boards, which use cards for components of smart things, and an app for rapidly creating interactive prototypes of ideas of smart things (Fig. 1, 2, 3 and 4). Boards also employ dedicated cards and plug-ins for enabling learners to reflect on what they do along three main themes—sustainability for culture and people, sustainability for the environment, besides usability, feasibility, and creativity (Fig. 5). In the art class, the tool was used in 5 days of work. With the first board, learners elaborated a story, co-creating part of the context, and imagined initial ideas of smart things. Learners evolved their initial ideas and reflected on them by using the other two boards of IoTgo, with its app, reflection cards, and plug-ins.

Data were collected in relation to the evolution of ideas and reflections, both from the perspective of teachers, with diaries and interviews, and learners, through the filled-in IoTgo boards. According to the data analysis reported above, during the work in class, all initial ideas matured and became feasible to turn into working

prototypes. According to the teacher, rounds of reflections and revisions were needed for that, and the “structure” of the boards made sure all learners could achieve it. This is also confirmed by the frequency of reflection cards used in relation to usability or feasibility itself, which was the highest among all reflection cards used by learners (50%, i.e., 16 out of 32).

All final ideas followed similar patterns, probably due to the role of the IoTgo app, which generated programs following similar behaviours. At the same time, according to the teacher, the app and reflection questions enabled learners to quickly deploy and test their ideas, thereby triggering rounds of necessary revisions and a deeper understanding of how physical-computing devices work and how to employ them in their smart-thing ideas. Interestingly, besides reflections concerning usability and feasibility, most reflections by learners were related to the environment and several sustainability goals of the 2030 Agenda (31%, i.e., 10 out of 32) [14]. This choice may have been stirred by the initial story excerpt or teens’ sensitivity to the topic itself.

This paper thus offers a description of the IoTgo tool and a case study. This shows how the tool enabled a teacher of an art subject to bring physical computing into her high-school class and promote reflections along the process, on her own. The current analysis is limited to the evolution of ideas of teens and the frequency of reflections per theme. Ongoing work is considering the programs that learners also developed as further means to gain insight into the evolution of their ideas and computing skills, besides a thematic analysis of answers to reflection questions and learners’ reflection questions. Future work will consider the case study with teachers of a novel training program concerning physical computing for generalising it to other classes, and for assessing the evolution of the computing skills of their classes in a systematic manner.

REFERENCES

- [1] Charoula Angeli and Michail Giannakos. 2020. Computational thinking education: Issues and challenges. *Computers in Human Behavior* 105 (2020), 106185. <https://doi.org/10.1016/j.chb.2019.106185>
- [2] Andrea Bonani, Rosella Gennari, Alessandra Melonio, and Mehdi Rizvi. 2023. Design and Computational Thinking with IoTgo: What Teachers Think. In *Methodologies and Intelligent Systems for Technology Enhanced Learning, 12th International Conference*, Marco Temperini, Vittorio Scarano, Ivana Marenzi, Milos Kravcik, Elvira Popescu, Rosa Lanzilotti, Rosella Gennari, Fernando De La Prieta, Tania Di Mascio, and Pierpaolo Vittorini (Eds.). Springer International Publishing, Cham, 165–174.
- [3] John W. Creswell. 2014. *Research Design*. SAGE.
- [4] Dries De Roeck, Jürgen Tanghe, Alexis Jacoby, Ingrid Moons, and Karin Slegers. 2019. Ideas of Things: The IOT Design Kit. In *Companion Publication of the 2019 on Designing Interactive Systems Conference 2019 Companion* (San Diego, CA, USA) (DIS '19 Companion). Association for Computing Machinery, New York, NY, USA, 159–163. <https://doi.org/10.1145/3301019.3323888>
- [5] Rosella Gennari, Maristella Matera, Alessandra Melonio, Mehdi Rizvi, and Eftychia Roumelioti. 2021. Reflection and awareness in the design process: children ideating, programming and prototyping smart objects. *Multimedia Tools and Applications* 80, 26–27 (2021), 34909 – 34932.
- [6] Rosella Gennari, Maristella Matera, Alessandra Melonio, Mehdi Rizvi, and Eftychia Roumelioti. 2022. The evolution of a toolkit for smart-thing design with children through action research. *International Journal of Child-Computer Interaction* 31 (2022), 100359. <https://doi.org/10.1016/j.ijcci.2021.100359>
- [7] Rosella Gennari and Mehdi Rizvi. 2021. At the Frontiers of Art and IoT: The IoTgo Toolkit as a Probe for Artists. In *CHIItaly 2021: 14th Biannual Conference of the Italian SIGCHI Chapter* (Bolzano, Italy) (CHIItaly '21). Association for Computing Machinery, New York, NY, USA, Article 16, 4 pages. <https://doi.org/10.1145/3464385.3464727>
- [8] Michail N. Giannakos and Letizia Jaccheri. 2013. What Motivates Children to Become Creators of Digital Enriched Artifacts?. In *Proceedings of the 9th ACM Conference on Creativity & Cognition* (Sydney, Australia). Association for

- Computing Machinery, New York, NY, USA, 104–113. <https://doi.org/10.1145/2466627.2466634>
- [9] Danielle Herro, Cassie Quigley, Holly Plank, Oluwadara Abimbade, and Aileen Owens. 2022. Instructional practices promoting computational thinking in STEAM elementary classrooms. *Journal of Digital Learning in Teacher Education* 38, 4 (2022), 158–172. <https://doi.org/10.1080/21532974.2022.2087125> arXiv:<https://doi.org/10.1080/21532974.2022.2087125>
- [10] Marianne Kinnula, Netta Iivari, Iván Sánchez Milara, and Jani Ylioja. 2020. *Guidelines for Empowering Children to Make and Shape Digital Technology—Case Fab Lab Oulu*. Springer Singapore, Singapore, 153–177. https://doi.org/10.1007/978-981-15-6747-6_9
- [11] Micro:bit-Educational-Foundation. 2019. Micro:bit Educational Foundation | micro:bit. <https://microbit.org>. Accessed: 2019-09-06.
- [12] Simone Mora, Francesco Gianni, and Monica Divitini. 2017. Tiles: A Card-based Ideation Toolkit for the Internet of Things. In *Proceedings of the 2017 Conference on Designing Interactive Systems (Edinburgh, United Kingdom) (DIS '17)*. ACM, New York, NY, USA, 587–598. <https://doi.org/10.1145/3064663.3064699>
- [13] Marie-Monique Schaper, Rachel Charlotte Smith, Mariana Aki Tamashiro, Maarten Van Mechelen, Mille Skovhus Lunding, Karl-Emil Kjæer Bilstrup, Magnus Høholt Kaspersen, Kasper Løvborg Jensen, Marianne Graves Petersen, and Ole Sejer Iversen. 2022. Computational empowerment in practice: Scaffolding teenagers' learning about emerging technologies and their ethical and societal impact. *International Journal of Child-Computer Interaction* 34 (2022), 100537. <https://doi.org/10.1016/j.ijcci.2022.100537>
- [14] United Nations. 2022. The Sustainability Development Goals. <https://www.un.org/sustainabledevelopment/development-agenda/>